

# Exploring the Impact of Process Transparency on User Experience in AI Design Agents

Ailin Yu\*  
Alibaba Cloud Computing  
Hangzhou, China  
ailinyu68@gmail.com

Zehui Jin\*  
Alibaba Cloud Computing  
Hangzhou, China  
jinzehui75@gmail.com

Yan Zhu\*  
Alibaba Cloud Computing  
Hangzhou, China  
thenina@foxmail.com

Yue He\*  
Alibaba Cloud Computing  
Hangzhou, China  
heyue305@gmail.com

Tao Yang†  
Alibaba Cloud Computing  
Hangzhou, China  
komayangtao@gmail.com

Luping Wang†  
Alibaba Cloud Computing  
Hangzhou, China  
dopink@msn.com

## Abstract

As AI design agents increasingly incorporate chain-of-thought (CoT) reasoning into creative workflows such as poster generation, user understanding and trust in these complex systems have become critical. To explore how CoT outcomes can be effectively visualized, we define *process transparency* as the degree to which an AI agent reveals its reasoning and actions throughout task execution. We first conducted a formative study to identify 10 key information components within the execution process, which informed the development of high, medium, and low transparency levels. Our subsequent user study examined how process transparency influences user trust, satisfaction, and willingness to use AI design agents. The results showed that higher transparency significantly improved all three measures, highlighting the value of clear and actionable information. Our findings underscore the need for adaptive transparency mechanisms and offer design implications for building user-centered AI design agents in creative agentic workflows.

## CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; Empirical studies in interaction design; • **Computing methodologies** → Artificial intelligence.

## Keywords

AI design agents, chain of thought, user experience evaluation

### ACM Reference Format:

Ailin Yu, Zehui Jin, Yan Zhu, Yue He, Tao Yang, and Luping Wang. 2025. Exploring the Impact of Process Transparency on User Experience in AI Design Agents. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen,

\* Contributed equally to this research

† Co-corresponding authors: dopink@msn.com, komayangtao@gmail.com

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW Companion '25, Bergen, Norway

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3749256>

Norway. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3715070.3749256>

## 1 Introduction & Related Work

Advancements in reasoning, multimodality, and tool use have improved large language models like GPT-4o and Gemini 2.0 [27, 30], enabling them to handle complex tasks and giving rise to LLM-powered agents. These multimodal AI agents have impacted fields like design, allowing users to interact naturally and generate multimodal content, which enhances creativity and streamlines workflows [5, 22, 23, 29]. Meanwhile, commercial AI design agents like Figma Make [11], Lovart [20], and Runway [26] use multimodal AI for tasks like UI design, image generation, and content creation, allowing users with little design experience to create professional-quality outputs.

Chain-of-Thought (CoT) [33] and Chain-of-Action (CoA) [21] mechanisms have been proposed to enhance the reasoning capabilities of LLMs by incorporating logical steps and actions into interpretable sequences for building powerful agents [2, 14]. Prior studies have shown that these mechanisms make AI systems more understandable and trustworthy by making the reasoning process more transparent [2, 31, 32, 38]. However, how to effectively visualize the outcomes of CoT and CoA remains underexplored, particularly in AI design workflows handling complex visual design tasks, such as poster generation [18]. This limitation hampers transparency and underscores the need to address these issues to enhance the interpretability of AI systems and users' understanding when interacting with AI agents [9, 28, 36, 39].

In this study, we define process transparency as the extent to which an AI Design Agent discloses its reasoning process during key stages of task execution, including planning, execution, and summarization [7, 13, 18, 25]. We focus on addressing the following research questions: (1) **What design knowledge should be included in the Chain-of-Thought (CoT) and Chain-of-Action (CoA) processes?** (2) **How do varying levels of process transparency in the reasoning process influence user trust, satisfaction, and willingness to use the system?** To explore these questions, we first conducted a formative study with six professional UX designers to identify the critical information that should be presented to users. Drawing on these insights, we designed three levels of transparency to simulate interactions with an AI Design Agent performing a poster generation task. Subsequently, we

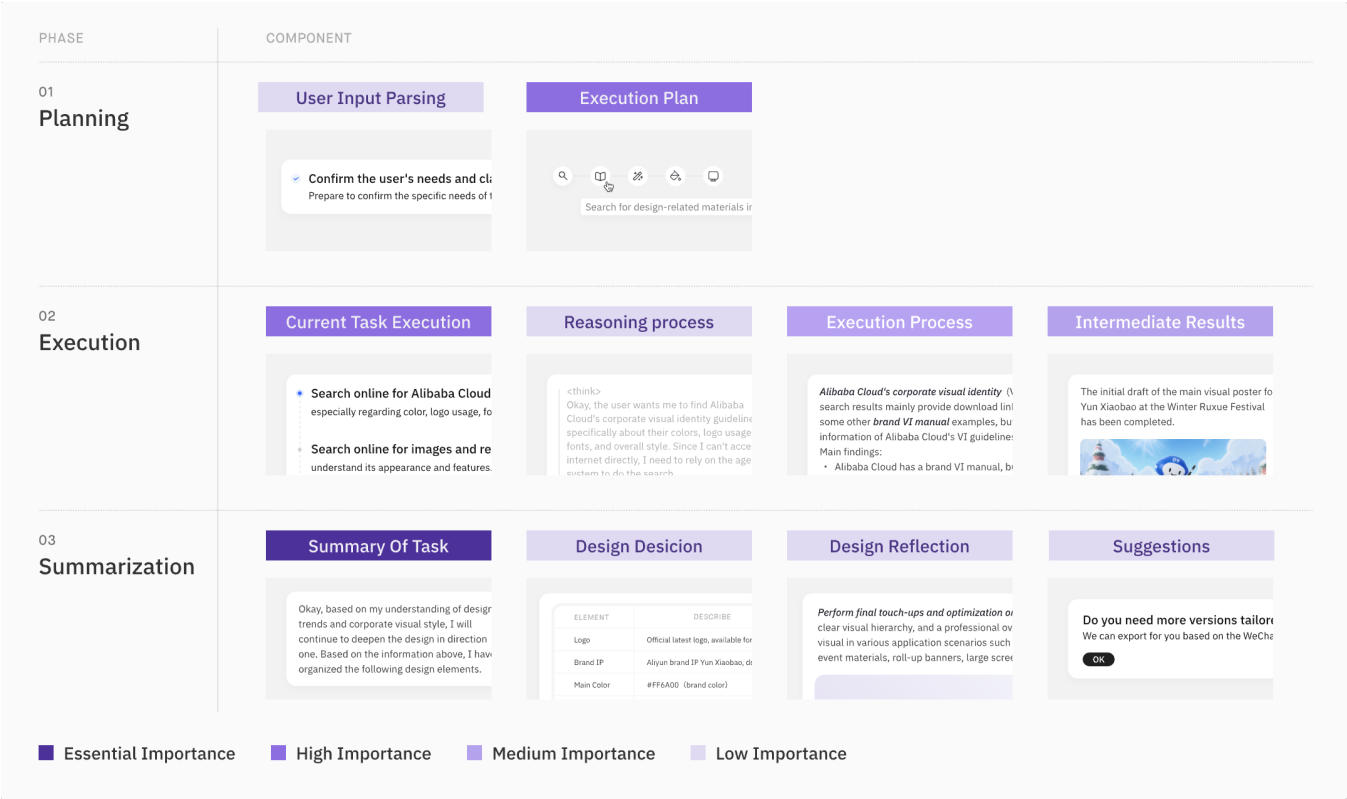


Figure 1: Critical information identified with varying levels of importance during the task execution of an AI agent.

conducted a user study with 40 participants, revealing that higher levels of transparency significantly enhanced user trust, satisfaction, and the intention to use the system. Based on these findings, we propose design implications that highlight the importance of process transparency in AI design agents. Our results contribute valuable insights for designing agentic systems that foster improved usability and trustworthiness.

2 Formative Study

We conducted a formative study to define process transparency in AI design agents and identify the relevant information to be displayed in the Chain-of-Thought (CoT) and Chain-of-Action (CoA) structures. Drawing on research in AI decision-making frameworks [1, 6, 10], explainable AI (XAI) [13, 19, 25], and Human-AI Interaction [3, 17, 37], we conceptualized process transparency and divided it into three distinct phases: *Planning*, which involves analyzing user needs, breaking down tasks, and setting goals; *Execution*, which includes the AI design agent’s operations during task execution, such as reasoning, resource acquisition, and generating intermediate results; and *Summarization*, which focuses on reviewing task outcomes, explaining design decisions, and providing feedback for optimization.

We then conducted expert interviews and brainstorming sessions with six professional UX designers (mean experience=6.5 years, SD=2.57) to formulate and refine the information dimensions

identified in the previous phase. The experts were recruited internally from within the company. These discussions also incorporated practical design contexts in poster generation, which helped us prioritize the most relevant information for each phase. As a result, we identified 10 key information components across the three phases, presented in Figure 1. To further clarify the role of reasoning and action in each stage, we aligned them with CoT and CoA mechanisms: the planning phase reflects CoT-driven reasoning, where the agent forms internal interpretations and strategies; the execution phase involves a combination of CoT and CoA, as the agent iteratively reasons and performs actions; and the summarization phase aligns with CoA, focusing on consolidating and reflecting on completed operations. Specifically, the planning phase includes user input parsing and plan execution; the execution phase includes the current task in progress, reasoning process, execution steps, and intermediate results; and the summarization phase includes a summary of the task, design decisions, design reflections, and further suggestions.

3 User Study

To investigate whether process transparency influences users’ evaluations of AI design agents—specifically *trust*, *satisfaction*, and *willingness to use* [15, 25, 28, 32, 35]—and to explore the optimal level of transparency for informing the design and reasoning process of such agents, we conducted a within-subjects experiment using a three-level transparency condition (high, medium, low). We

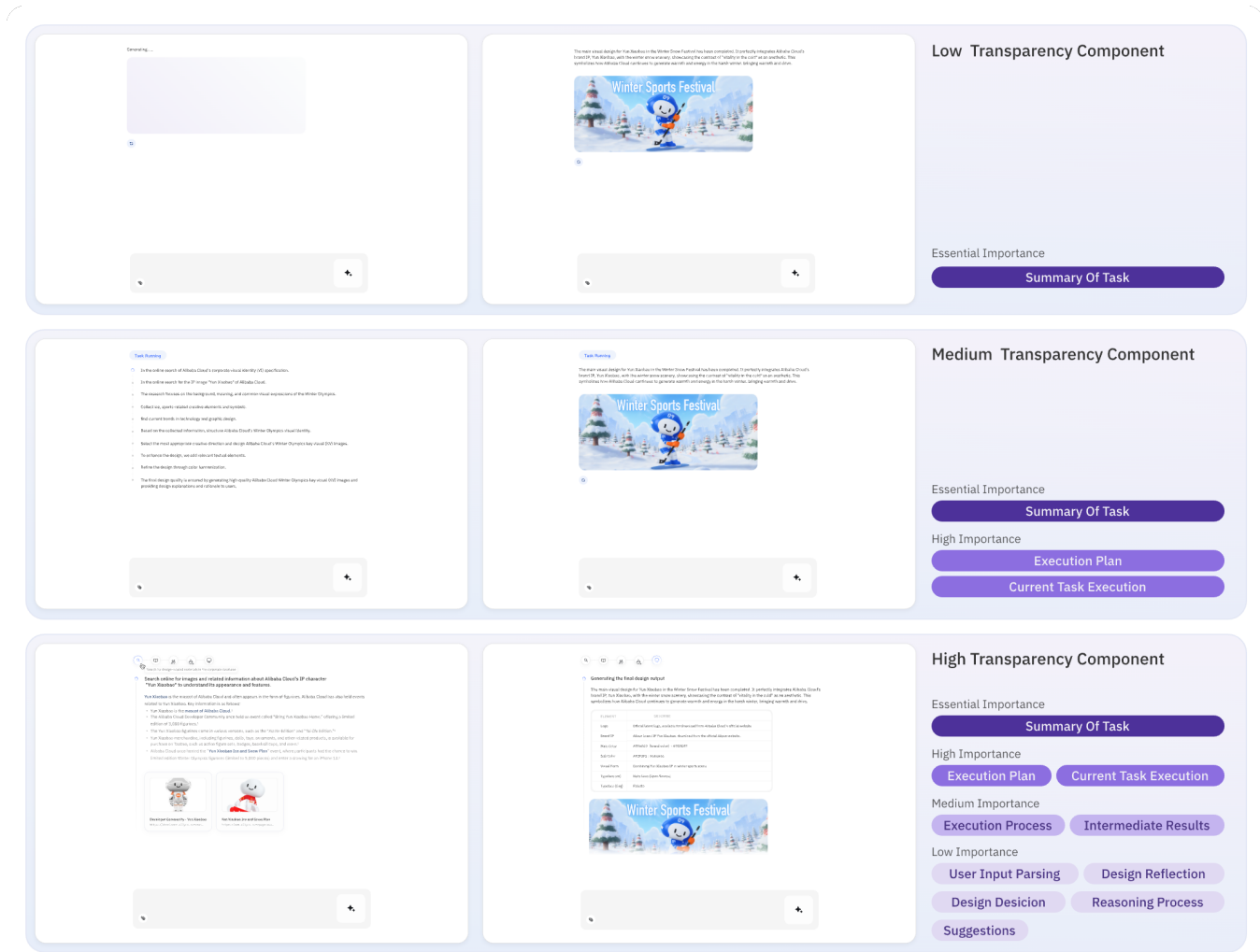


Figure 2: Examples of Level of Process Transparency

created three representative examples and carried out both experimental studies and follow-up focus group interviews to gain deeper insights into user experience [4, 8].

### 3.1 Different Level of Process Transparency

Based on our formative study, we proposed three levels of transparency with different information components and created corresponding simulation materials for a poster generation task, presented in Figure 2. **High transparency** included all identified components; **medium transparency** included essential and high-importance components, such as the execution plan, current task execution, and summary of the task; and **low transparency** included only the essential component, namely the summary of the task. This setup allowed us to systematically investigate how the amount of process information presented to users affects their overall user experience.

### 3.2 Participants

We recruited 40 participants (aged 19–35; 77.5% female), including 33 students with a design background and 7 professional designers. Participants were recruited through two channels: design students were invited via online outreach from a university program, while professional designers were recruited internally within the company. These professional designers were not involved in the earlier formative study. All participants had prior experience with AI-based image generation tools, such as GPT-4o, Midjourney, Stable Diffusion, or Chinese platforms like Doubao. All data were anonymized and used solely for academic research. Participants gave their informed consent before the study and participated voluntarily without compensation.

### 3.3 Procedure

In our experiment, process transparency was manipulated as an independent variable with three levels (high, medium, low) using

**Table 1: Descriptive statistics (Mean  $\pm$  SD) across different transparency conditions (N = 40, 7-point Likert scale)**

| Variables          | High Transparency   | Medium Transparency | Low Transparency    |
|--------------------|---------------------|---------------------|---------------------|
| Trust              | 5.14 ( $\pm 1.08$ ) | 4.88 ( $\pm 1.13$ ) | 4.14 ( $\pm 1.65$ ) |
| Satisfaction       | 5.33 ( $\pm 1.31$ ) | 4.95 ( $\pm 1.38$ ) | 3.98 ( $\pm 1.95$ ) |
| Willingness to Use | 5.23 ( $\pm 1.21$ ) | 4.83 ( $\pm 1.43$ ) | 4.03 ( $\pm 1.80$ ) |

a within-subjects design. Trust, satisfaction, and willingness to use were the dependent variables, measured via a 7-point Likert scale (1=strongly disagree, 7=strongly agree) [12, 24]. Trust was measured using the TOAST scale (Trust of Automated Systems Test) [34]. Participants watched three two-minute simulated interaction videos, each depicting the AI design agent completing a poster generation task with varying transparency levels. After each video, participants filled out a questionnaire to evaluate trust, satisfaction, and willingness to use the agent. The procedure included three trials with counterbalanced order of transparency conditions using a Latin square design. Demographic information was collected, and focus group discussions were held afterward for qualitative feedback.

## 4 Results

To examine the effect of process transparency on user experience, we conducted a one-way repeated measures ANOVA on three key outcome variables: user trust, user satisfaction, and willingness to use. Mauchly's test indicated that the assumption of sphericity was violated for all three analyses, so Greenhouse-Geisser corrections were applied. We observed consistent effects of the transparency level in all three outcome variables: trust, satisfaction, and willingness to use. We report our results in Table 1.

**Trust** The analysis revealed a significant main effect of transparency on trust ( $F(1.346, 52.502) = 14.47, p < .001$ , partial  $\eta^2 = .271$ ). Pairwise comparisons showed that trust was significantly higher under high transparency ( $M = 5.14, SD = 1.08$ ) compared to both medium transparency ( $M = 4.88, SD = 1.13, p = .026$ ) and low transparency ( $M = 4.14, SD = 1.65, p < .001$ ). The difference between medium and low transparency was also significant ( $p = .001$ ). These results suggest that transparency is a key factor in fostering user trust.

**Satisfaction** The analysis similarly showed a significant main effect of transparency on satisfaction ( $F(1.553, 60.577) = 16.036, p < .001$ , partial  $\eta^2 = .291$ ). Satisfaction was significantly higher under high transparency ( $M = 5.33, SD = 1.31$ ) compared to both medium transparency ( $M = 4.95, SD = 1.38, p = .045$ ) and low transparency ( $M = 3.98, SD = 1.95, p < .001$ ). The difference between medium and low transparency was also significant ( $p < .001$ ). This suggests that higher transparency in the generative process significantly enhances user satisfaction.

**Willingness to use** The analysis revealed a significant main effect of transparency on willingness to use,  $F(1.454, 56.690) = 15.167, p < .001$ , partial  $\eta^2 = .280$ . Willingness to use was significantly higher under high transparency ( $M = 5.23, SD = 1.21$ ) compared to medium transparency ( $M = 4.83, SD = 1.43, p = .016$ ) and low transparency ( $M = 4.03, SD = 1.80, p < .001$ ). The difference between medium and

low transparency was also significant ( $p = .001$ ). These findings further highlight the critical role of transparency in motivating users to engage with the system.

**User Feedback** To complement the quantitative findings, we conducted a descriptive analysis of questionnaire responses on satisfaction preferences. In addition, four researchers independently documented and analyzed the content of two focus group sessions. Select open-ended responses from participants were used as illustrative examples to enrich interpretation. Among the 40 participants, 26 rated the high transparency interface as the most satisfying condition, commonly noting that it helped them understand system logic and refine their inputs. In two of the four focus groups, participants explicitly mentioned a sense of control. As two experienced designers noted, *"If the explanations are sufficiently detailed and logical, even suboptimal outputs can help me identify issues and optimize inputs to achieve better results."* Additionally, 12 students preferred medium transparency and rated it highest in terms of satisfaction. Some students remarked, *"The reasoning process provided by high transparency is too verbose and irrelevant—I only need to focus on the task summary and the final outcomes."*

To summarize, our study demonstrates that transparency plays an irreplaceable role in shaping user trust, satisfaction, and willingness to use. Specifically, users reported higher levels of trust, satisfaction, and willingness to use under high transparency conditions, whereas these ratings were significantly lower under low transparency conditions. The medium transparency condition showed intermediate levels. Further pairwise comparisons indicated that the differences between high, medium, and low transparency conditions were statistically significant across all three variables.

## 5 Discussion

Our quantitative analysis revealed a significant main effect of process transparency. Specifically, providing more detailed information during the poster generation process resulted in higher user ratings of trust, satisfaction, and willingness to use. The high-transparency condition consistently received more favorable evaluations. Complementing these findings, qualitative data from post-task interviews revealed that participants felt more informed, in control, and confident in the agent's decisions when given access to detailed reasoning and action steps. **Enhanced understanding arises when transparency makes the system's logic visible** High-transparency conditions enabled users to follow the AI's reasoning by revealing intermediate steps. Many participants noted that this process visibility offered clear insight into the system's design rationale and decision-making logic, thereby increasing their trust in the system. This finding aligns with prior work by Kim et al., which demonstrated that users prefer AI systems that explain

their decisions and support user feedback, rather than those that provide opaque, one-way outputs [16].

**Transparency supports perceived agency** Participants reported feeling more confident and in control when they could “see what the system was thinking and doing.” In contrast, under low-transparency conditions, the lack of information led to confusion and frustration, particularly when outputs were unsatisfactory, as users were unable to trace the source of errors. Although this sense of control was not literal, it significantly shaped users’ trust in the AI agent. This perceived agency can be further interpreted through the lens of CoT and CoA: users felt empowered when they could trace the agent’s reasoning process (CoT) and observe its concrete steps during execution (CoA), especially when debugging or reflecting on suboptimal outcomes.

**Providing Relevant and Useful Affordances** While many participants found the high-transparency condition helpful, some felt that details like the reasoning or execution process were unnecessary and preferred to focus on the final output to save time—“just show me the referenced image,” one user commented. Others noted that explanations of common-sense or generic knowledge made the system feel less intelligent, and said they valued more specific, need-relevant information instead. Meanwhile, several participants wanted clearer progress indicators and estimated task duration to build trust and manage expectations. Overall, feedback from the focus group aligned with the formative study’s rankings of transparency components—except for intermediate steps, which were viewed as more important than experts had anticipated, nearly on par with task summaries.

In summary, process transparency enhances more than just information visibility—it also psychologically empowers users by fostering trust and a sense of agency. Systems that can explain themselves tend to elicit more positive feedback, whereas opaque systems often lead to uncertainty and frustration, particularly when generating unexpected results. These findings help explain why high-transparency conditions consistently result in more favorable user evaluations.

## 6 Design Implications

We identified three key implications for implementing process transparency in AI design agents.

**Intermediate Results in the Execution Phase** Many participants expressed a strong preference for viewing intermediate outputs during task execution, as this helped them identify misalignments early and trace potential causes when outcomes deviated from expectations. In the case of *Manus*, intermediate results are displayed only during the summary phase, which may limit users’ ability to intervene in real time. By contrast, *Lovart* [20], a recently released design-focused AI tool, presents intermediate outputs during execution—a structure aligned with the high-transparency condition in our study. This practice highlights a promising direction for designing the timing of transparency in future AI design agents.

**Adjustable Transparency** Individual differences were observed in users’ preferences for transparency. While many users favored high transparency to better understand the system’s logic, others prioritized the final output and preferred to avoid process-related distractions. These findings suggest that future AI design tools

should support customizable transparency levels or present information in a flexible, structured format—such as through collapsible views.

**Task-Driven Adaptive Transparency** Complex tasks involving multi-step reasoning such as brand-guided design benefit from higher transparency, as it supports users’ understanding of system behavior. However, for simpler, well-defined tasks (e.g., text-to-image generation with clear prompts), lower transparency can improve efficiency by reducing unnecessary detail. As noted by several participants, excessive breakdowns of simple tasks led to frustration and diminished their perception of the system’s intelligence.

## 7 Conclusion & Future Work

This study investigated the key types of information involved in process transparency and how they affect users’ trust, satisfaction, and willingness to use AI design agents. The findings underscore the value of higher process transparency and inform a set of design implications aimed at making transparency more adaptive, actionable, and user-controllable. However, the study has limitations. First, the sample was skewed toward design-background participants (81% were students or professionals in design), which may limit generalizability to broader user groups such as marketers or non-design professionals. Second, the use of pre-scripted demonstration videos ensured experimental control but reduced ecological validity, as it did not fully capture the dynamics of real-time user-agent interaction. Future work may explore live interaction settings and more diverse user populations, as well as develop methods to dynamically model and visualize CoT and CoA structures based on user intent and task complexity, in order to better support adaptive transparency in real-time AI systems.

## References

- [1] Deepak Bhaskar Acharya, Karthigeyan Kuppan, and B Divya. 2025. Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey. *IEEE Access* (2025).
- [2] Ahsan Ali and Zhao Feng. 2023. Layered Chain-of-Thought Prompting: A Step Towards More Transparent AI Models. (2023).
- [3] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, et al. 2019. Guidelines for human-AI interaction. In *Proceedings of the 2019 chi conference on human factors in computing systems*. 1–13.
- [4] Ann Blandford, Dominic Furniss, and Stephann Makri. 2016. *Qualitative HCI research: Going behind the scenes*. Morgan & Claypool Publishers.
- [5] Minsuk Chang, Stefania Druga, Alexander J Fiannaca, Pedro Vergani, Chinmay Kulkarni, Carrie J Cai, and Michael Terry. 2023. The prompt artists. In *Proceedings of the 15th Conference on Creativity and Cognition*. 75–87.
- [6] Chhavi Chawla, Siddharth Chatterjee, Sanketh Siddanna Gadadinni, Pulkit Verma, and Sourav Banerjee. 2024. Agentic AI: The building blocks of sophisticated AI business applications. *Journal of AI, Robotics & Workplace Automation* 3, 3 (2024), 1–15.
- [7] Jessie YC Chen, Shan G Lakhmani, Kimberly Stowers, Anthony R Selkowitz, Julia L Wright, and Michael Barnes. 2018. Situation awareness-based agent transparency and human-autonomy teaming effectiveness. *Theoretical issues in ergonomics science* 19, 3 (2018), 259–282.
- [8] John W Creswell and Vicki L Plano Clark. 2017. *Designing and conducting mixed methods research*. Sage publications.
- [9] Stephan Diederich, Alfred Benedikt Brendel, Stefan Morana, and Lutz Kolbe. 2022. On the design of and interaction with conversational agents: An organizing and assessing review of human-computer interaction research. *Journal of the Association for Information Systems* 23, 1 (2022), 96–138.
- [10] Zane Durante, Qiuyuan Huang, Naoki Wake, Ran Gong, Jae Sung Park, Bidipta Sarkar, Rohan Taori, Yusuke Noda, Demetri Terzopoulos, Yejin Choi, et al. 2024. Agent ai: Surveying the horizons of multimodal interaction. *arXiv preprint arXiv:2401.03568* (2024).

- [11] Figma 2025. *Introducing Figma Make: A new way to test, edit, and prompt designs*. <https://www.figma.com/blog/introducing-figma-make/>
- [12] Craig Finstad. 2010. Response interpolation and scale sensitivity: Evidence against 5-point scales. *Journal of usability studies* 5, 3 (2010), 104–110.
- [13] David Gunning and David Aha. 2019. DARPA's explainable artificial intelligence (XAI) program. *AI magazine* 40, 2 (2019), 44–58.
- [14] Ziyu Guo, Renrui Zhang, Chengzhuo Tong, Zhizheng Zhao, Peng Gao, Hongsheng Li, and Pheng-Ann Heng. 2025. Can We Generate Images with CoT? Let's Verify and Reinforce Image Generation Step by Step. *arXiv preprint arXiv:2501.13926* (2025).
- [15] Sage Kelly, Sherrie-Anne Kaye, and Oscar Oviedo-Trespalacios. 2023. What factors contribute to the acceptance of artificial intelligence? A systematic review. *Telematics and Informatics* 77 (2023), 101925.
- [16] Soomin Kim, Jinsu Eun, Changhoon Oh, and Joonhwan Lee. 2025. "Journey of finding the best query": Understanding the user experience of AI image generation system. *International Journal of Human-Computer Interaction* 41, 2 (2025), 951–969.
- [17] Tianyi Li, Mihaela Vorvoreanu, Derek DeBellis, and Saleema Amershi. 2023. Assessing human-AI interaction early through factorial surveys: a study on the guidelines for human-AI interaction. *ACM Transactions on Computer-Human Interaction* 30, 5 (2023), 1–45.
- [18] Q Vera Liao, Hariharan Subramonyam, Jennifer Wang, and Jennifer Wortman Vaughan. 2023. Designerly understanding: Information needs for model transparency to support design ideation for AI-powered user experience. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–21.
- [19] Brian Y Lim, Anind K Dey, and Daniel Avrahami. 2009. Why and why not explanations improve the intelligibility of context-aware intelligent systems. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 2119–2128.
- [20] Lovart 2025. *Transform prompts into masterpieces with Lovart, the revolutionary AI design agent*. <https://www.lovart.ai/>
- [21] Zhenyu Pan, Haozheng Luo, Manling Li, and Han Liu. 2024. Chain-of-action: Faithful and multimodal question answering through large language models. *arXiv preprint arXiv:2403.17359* (2024).
- [22] Kaushikkumar Patel, Divya Beeram, Prashanthi Ramamurthy, Prerak Garg, and Sandeep Kumar. 2024. Ai-enhanced design: revolutionizing methodologies and workflows. *Development (TJAIRD)* 2, 1 (2024), 135–157.
- [23] Andrew Pearson. 2023. The rise of CreAltives: Using AI to enable and speed up the creative process. *Journal of AI, Robotics & Workplace Automation* 2, 2 (2023), 101–114.
- [24] Carolyn C Preston and Andrew M Colman. 2000. Optimal number of response categories in rating scales: reliability, validity, discriminating power, and respondent preferences. *Acta psychologica* 104, 1 (2000), 1–15.
- [25] Yao Rong, Tobias Leemann, Thai-Trang Nguyen, Lisa Fiedler, Peizhu Qian, Vaibhav Unhelkar, Tina Seidel, Gjergji Kasneci, and Enkelejda Kasneci. 2023. Towards human-centered explainable ai: A survey of user studies for model explanations. *IEEE transactions on pattern analysis and machine intelligence* 46, 4 (2023), 2104–2122.
- [26] Ketan Suhaas Saichandran, Xavier Thomas, Prakhar Kaushik, and Deepti Ghadiyaram. 2025. Progressive Prompt Detailing for Improved Alignment in Text-to-Image Generative Models. *arXiv preprint arXiv:2503.17794* (2025).
- [27] Sakib Shahriar, Brady D Lund, Nishith Reddy Mannuru, Muhammad Arbab Arshad, Kadhim Hayawi, Ravi Varma Kumar Bevara, Aashrith Mannuru, and Laiba Batool. 2024. Putting gpt-4o to the sword: A comprehensive evaluation of language, vision, speech, and multimodal proficiency. *Applied Sciences* 14, 17 (2024), 7782.
- [28] Yonadav Shavit, Sandhini Agarwal, Miles Brundage, Steven Adler, Cullen O'Keefe, Rosie Campbell, Teddy Lee, Pamela Mishkin, Tyna Eloundou, Alan Hickey, et al. 2023. Practices for governing agentic AI systems. *Research Paper, OpenAI* (2023).
- [29] Yuying Tang, Ningning Zhang, Mariana Cancia, and Zhigang Wang. 2024. Exploring the Impact of AI-generated Image Tools on Professional and Non-professional Users in the Art and Design Fields. In *Companion Publication of the 2024 Conference on Computer-Supported Cooperative Work and Social Computing*. 451–458.
- [30] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805* (2023).
- [31] Eric S Vorm and David JY Combs. 2022. Integrating transparency, trust, and acceptance: The intelligent systems technology acceptance model (ISTAM). *International Journal of Human-Computer Interaction* 38, 18–20 (2022), 1828–1845.
- [32] Yanyun Wang, Xumei Fang, Zan Xu, Jianye Li, and Luping Wang. 2025. Exploring the Impact of Explainability in Large Language Model (LLM) Applications on User Experience. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–8.
- [33] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.
- [34] Heather M Wojton, Daniel Porter, Stephanie T. Lane, Chad Bieber, and Poornima Madhavan. 2020. Initial validation of the trust of automated systems test (TOAST). *The Journal of social psychology* 160, 6 (2020), 735–750.
- [35] Wentao Wu, Ben Zhang, Shuting Li, and Hehai Liu. 2022. Exploring factors of the willingness to accept AI-assisted learning environments: an empirical investigation based on the UTAUT model and perceived risk theory. *Frontiers in Psychology* 13 (2022), 870777.
- [36] Wei Xu, Marvin J Dainoff, Liezhong Ge, and Zaifeng Gao. 2023. Transitioning to human interaction with AI systems: New challenges and opportunities for HCI professionals to enable human-centered AI. *International Journal of Human-Computer Interaction* 39, 3 (2023), 494–518.
- [37] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining whether, why, and how human-AI interaction is uniquely difficult to design. In *Proceedings of the 2020 chi conference on human factors in computing systems*. 1–13.
- [38] Zhibin Zhou, Zhuoshu Li, Yuyang Zhang, and Lingyun Sun. 2022. Transparent-AI blueprint: Developing a conceptual tool to support the design of transparent AI agents. *International Journal of Human-Computer Interaction* 38, 18–20 (2022), 1846–1873.
- [39] Jichen Zhu, Antonios Liapis, Sebastian Risi, Rafael Bidarra, and G Michael Youngblood. 2018. Explainable AI for designers: A human-centered perspective on mixed-initiative co-creation. In *2018 IEEE conference on computational intelligence and games (CIG)*. IEEE, 1–8.